

Katrina Engelsted

GIS CANDIDATE · CITY OF BOULDER — PLANNING & DEVELOPMENT SERVICES

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CORE EXPERTISE

GIS & Spatial Analysis

- Spatial analysis & modeling
- Geodatabase design
- Cartographic design (print & digital)
- Spatial statistics
- ArcGIS

Databases & Big Data

- Relational & geospatial databases
- SQL
- Large-scale data analysis
- BigQuery
- Data quality & validation

Mobile & Field Data Collection

- Fulcrum
- Survey123
- Field data workflows
- Asset-management integration

Workflow Automation

- Python scripting
- Process automation
- Reporting pipelines
- Technical troubleshooting

Collaboration & Public Service

- Cross-departmental coordination
- Customer service
- Federal & public-sector GIS
- Metadata & documentation

EDUCATION

BA, Geography (GIS), Honors

Colgate University 2011
Honors thesis: spatial-statistics analysis using kriging, IDW, and local Moran's I.

MBA

Quantic School of Business and Technology 2021

SUMMARY

GIS professional with a **BA in Geography (GIS), Honors**, who solves problems with data at scale — relational and geospatial databases, large record sets, and the mobile field-data collection and automated workflows that keep that data usable. I've designed geodatabases and mapping tools in **ArcGIS**, worked hands-on with **mobile field data collection (Fulcrum, Survey123)** connecting GIS to real-world asset management, and built automated data and reporting workflows across multiple organizations (Zayo, Upstream AI, Sunlight). I write **Python** for process automation, work with relational databases and SQL at enterprise scale, and consistently serve as the translator between technical data work and the field staff, planners, or customers who rely on it. Based in Boulder, and excited to bring that background home to the city's Planning & Development Services team.

PROFESSIONAL EXPERIENCE

- Founding Technical Advisor** — *Upstream AI* June 2026 — Present
 - Build **automated workflows** (AI-agent workflows, RAG pipelines) that reduce manual, repetitive technical work for non-technical end users
- Business Data Analyst** — *Sunlight Inc.* June 2023 — June 2025
 - Automated inventory, pricing, and sales-reporting workflows for a small retail business, turning raw data into decisions owners could act on week to week
- Senior Technical Consultant** — *Lytics* 2022 — 2023
 - Solved a **large-scale data problem** for an enterprise client (tens of millions of records): used SQL, BigQuery, Python, and APIs to validate, model, and resolve a **relational database** into a usable identity model
 - Documented data architecture and standards in a playbook used across future engagements
- Marketing Automation Manager** — *Zayo Group* 2022
 - Automated** lead-scoring and reporting workflows end-to-end, turning a database of raw lead data into a system sales could act on, lifting conversion 15%
- Head of Marketing Technology** — *Teren, Inc. (formerly SolSpec)* 2018 — 2021
 - Worked within a **geospatial-risk company performing spatial analysis and modeling**.
 - Worked with **large geospatial datasets** — machine-learning models built from drone and satellite imagery — to assess environmental and infrastructure risk, turning big, messy geospatial data into decision-ready information
 - Partnered across technical and non-technical teams to translate geospatial analysis for institutional customers
- Solutions Engineer** — *Fulcrum* June 2015 — January 2017
 - Mobile field data-collection platform for GIS and asset management.*
 - Helped clients integrate **mobile field data collection** into enterprise GIS and asset-management workflows, connecting data collected in the field directly to backend databases and dashboards
 - Built and maintained web applications and custom dashboards for querying, updating, visualizing, analyzing, and **mapping geographic data**
 - Led product testing, API documentation, and user guides for the mobile data-collection platform
- Solutions Engineer** — *Homebot* January 2017 — March 2018
 - Designed a **database schema** for ingesting messy real-estate data from multiple sources at scale; translated technical findings for non-technical stakeholders
- Geospatial (GIS) Developer** — *National Park Service* 2012 — 2015
 - Built geospatial tools and cartographic products inside a federal land-management agency.*
 - Built and maintained **geodatabases** and web applications for querying, analyzing, and mapping **large geospatial datasets** in ArcGIS, Mapbox, and Carto; this work became **NPS Park Tiles**, the official basemap for park websites
 - Built wildfire-perimeter and endangered-species mapping tools for non-GIS park staff, maintaining data quality and documentation standards
- Early Career — Geospatial & Humanitarian Mapping**
 - Humanitarian OpenStreetMap Team · Cadasta (land-surveying nonprofit)*
 - Humanitarian OpenStreetMap Team** — QGIS-based disaster-response mapping and spatial data collection
 - Cadasta** — land and resource-rights registry documentation; evaluated and vetted field data-collection tools including **Survey123** and **Fulcrum** for land-surveying work